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#include <windows.h>
#include <GL/glut.h>
#include <cmath>
const float PI = 3.14159265358979323846;
float angle = 0.0f;
void drawCircle(float cx, float cy, float radius, int segments, bool
filled = false) {
    filled ? glBegin(GL_POLYGON) : glBegin(GL_LINE_LOOP);
    for (int i = 0; i < segments; ++i) {
        float theta = 2.0f * PI * i / segments;
        float x = radius * cosf(theta);
        float y = radius * sinf(theta);

        glVertex2f(cx + x, cy + y);
    }
    glEnd();
}

void display() {
    glClear(GL_COLOR_BUFFER_BIT);
    glColor3f(1.0, 0.0, 0.0);
    drawCircle(0.5f * cos(angle), 0.5f * sin(angle), 0.1f, 100);
    glColor3f(0.0, 1.0, 0.0);
    drawCircle(0.6f * sin(angle), 0.3f, 0.1f, 100);
    glColor3f(0.0, 0.0, 1.0);
    drawCircle(-0.6f, 0.4f * cos(angle), 0.1f, 100);
    glutSwapBuffers();
}

void update(int value) {
    angle += 0.01f;
    if (angle > 2 * PI) angle -= 2 * PI;
    glutPostRedisplay();
    glutTimerFunc(16, update, 0);
}

int main(int argc, char** argv) {
    glutInit(&argc, argv);
    glutInitDisplayMode(GLUT_DOUBLE | GLUT_RGB);
    glutInitWindowSize(600, 600);

    glutInitWindowPosition(100, 100);
    glutCreateWindow("Animated Circles");
    glutDisplayFunc(display);
    glutTimerFunc(0, update, 0);
    glutMainLoop();
    return 0;
}

```